

Cyber Security

Introduction

Cyber security refers to the practices, technologies, and processes designed to protect systems, networks, and data from unauthorized access, damage, or disruption. As organizations increasingly rely on digital infrastructure, cyber security has become a critical part of both personal and organizational risk management.

Common Threats

- Phishing: fraudulent messages designed to trick users into revealing credentials or installing malware.
- Malware: malicious software such as viruses, ransomware, and spyware.
- SQL Injection: exploiting improperly sanitized input to manipulate a database query.
- Cross-Site Scripting (XSS): injecting malicious scripts into content viewed by other users.
- Denial-of-Service (DoS): overwhelming a system with traffic to make it unavailable.

Authentication and Access Control

Strong authentication is a foundational security control. Multi-factor authentication combines something a user knows, such as a password, with something they have or are, such as a one-time code or biometric factor. Access control principles such as least privilege ensure that users and systems only have the permissions necessary to perform their intended function.

Data Protection

Sensitive data should be encrypted both at rest and in transit, using protocols such as TLS for network communication. Personally identifiable information and credentials require additional safeguards, such as masking or tokenization, so that sensitive values are not exposed in logs, error messages, or unauthorized parts of a system.

Secure Application Development

Building secure applications involves validating and sanitizing all user input, using parameterized queries to prevent injection attacks, keeping dependencies up to date to avoid known vulnerabilities, and following the principle of defense in depth, where multiple layers of controls protect a system even if one layer fails.

Network Security

Firewalls, intrusion detection systems, and network segmentation help limit an attacker's ability to move laterally within a network if one system is compromised. Regular patching and vulnerability scanning help identify and close security gaps before they can be exploited.

Incident Response

An effective incident response plan defines how an organization detects, contains, eradicates, and recovers from a security incident, along with clear communication procedures. Post-incident reviews help identify root causes and improve defenses against similar future incidents.

Employee Security Awareness

Since many breaches originate from human error, regular security awareness training helps employees recognize phishing attempts, use strong and unique passwords, and follow secure practices when handling company data, especially when working remotely or on personal devices.